

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

* * MECHANICAL PROPERTIES OF SOLIDS * *

LECTURE-01

PHYSICS

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Topics To Be Covered

- 1 Deforming Force
- 2 Restoring Force
- 3 Elasticity
- 4 Plasticity
- 5 Stress

* Hooker law
graph

* (Chapter 1-8) (Revision) Next Sunday 3 June 2026

* Half Marathon

* Attendance (15 days) Gift

* Quiz (Topper) Gift **

* 4 month (preparation)

$\Rightarrow$ Gift

$$\begin{array}{r} 496 \\ \hline 500 \end{array}$$

$$\begin{array}{r} 497 \\ \hline 500 \end{array}$$

$$\begin{array}{r} 498 \\ \hline 500 \end{array}$$

$$\begin{array}{r} 499 \\ \hline 500 \end{array}$$

$$\begin{array}{r} 500 \\ \hline 500 \end{array}$$

MECHANICAL PROPERTIES OF SOLIDS

Elasticity, Stress-strain relationship, Hooke's law, Young's modulus, bulk modulus, shear modulus of rigidity (qualitative idea only).
Poisson's ratio; elastic energy.

lec - 1

lec - 2

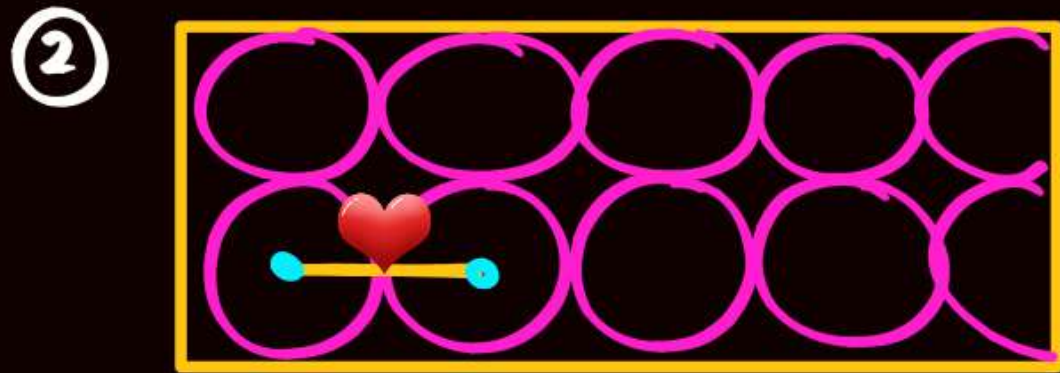
QNS PYQ + numerical

lecture no. 1

①

Properties of Solid (9th)

① State of matter



③ Inter-atomic distance
Small

④ Maximum Inter-atomic
force

JKBOSE
2016

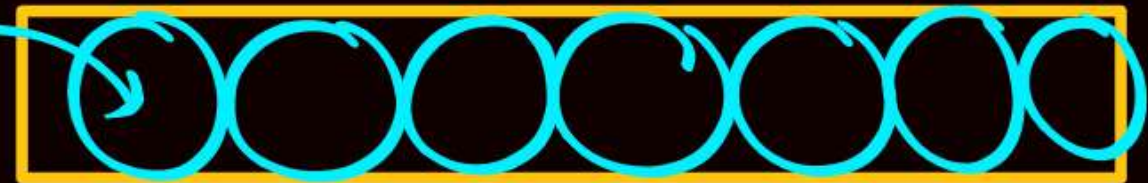
⑤ definite Shape

⑥ definite Volume

⑦ do not flow

⑧ Speed of Sound

Shazia



Steel → 6000 m/s

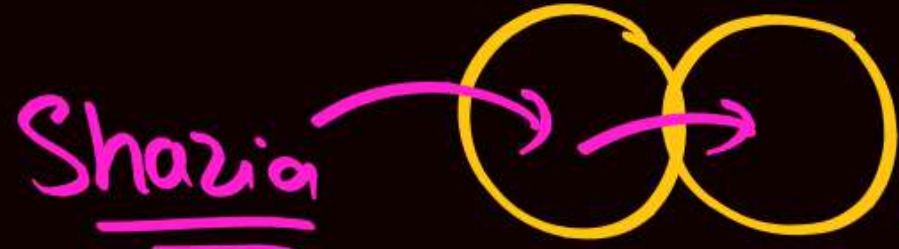
⑨ Density $\uparrow \rho = \frac{m}{V}$ (maximum)

⑩ Compressibility $\downarrow$
 $= 0$

2

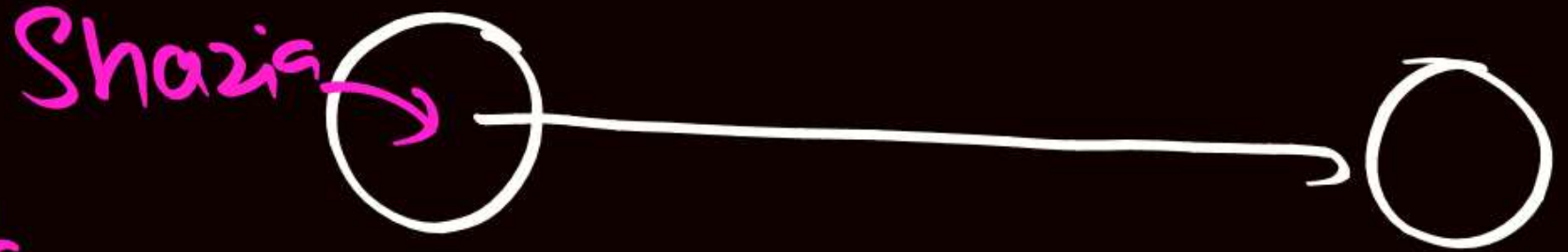
Solid

Atom Atom



Speed = 6000 m/s

liquid



Speed = 1500 m/s

Deforming Force

3



External force which change the shape of the body



F_{ext}

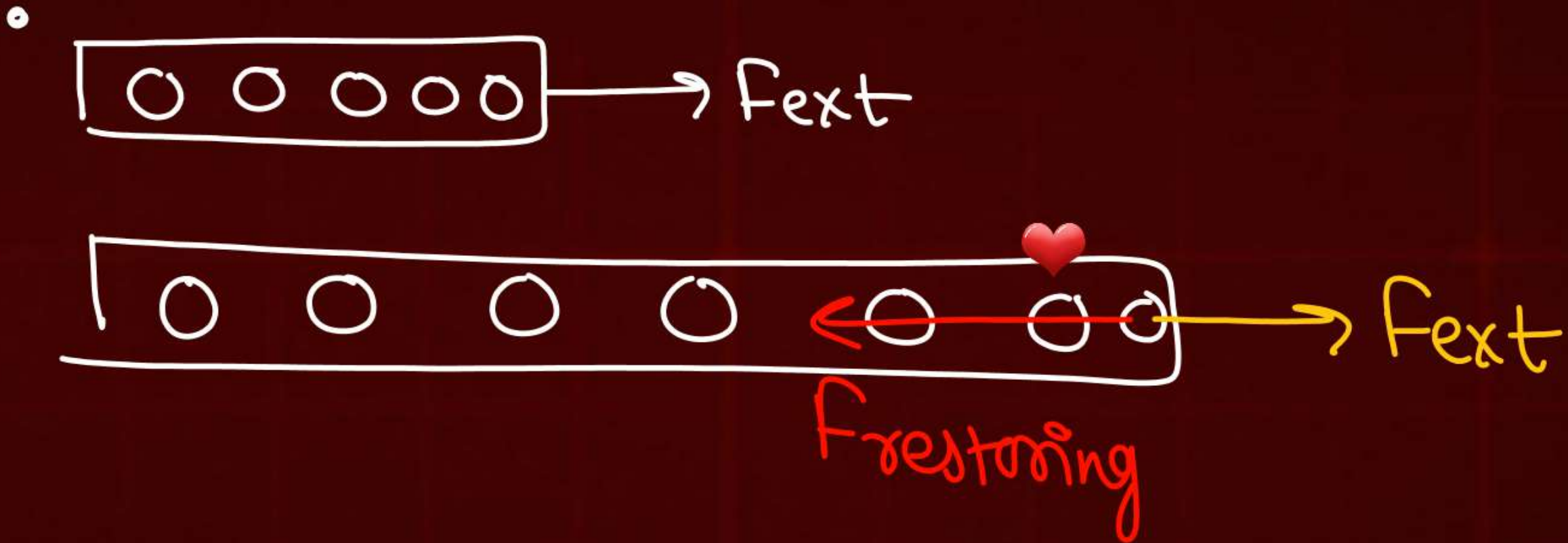
Deforming force

Restoring Force

4



Internal force which help the solid to regain the shape.



Elasticity

5



The property of solid to regain its shape ✓

After removing the Deforming force ✓



F_d

* Elasticity depend on
✓
X deforming force / Restoring force



F_R



Note one who regain its shape partly is more elastic

Q Which of the following more elastic? (Neet - 2019) ⑥

(a) Rubber

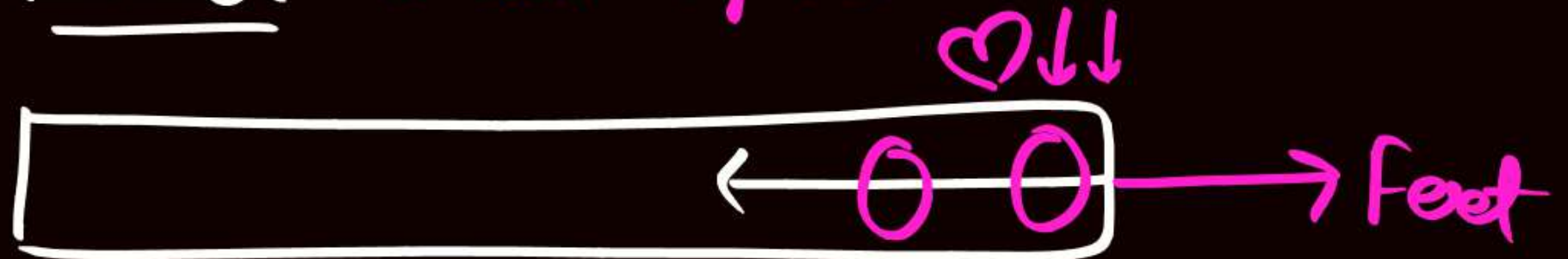
(b) Steel

① Steel ♡↑↑↑↑



Restoring force ↑↑↑↑↑↑↑↑

② Rubber Elasticity ↑↑↑↑↑



Restoring force ↓↓↓↓

Elasticity ↓↓

Plasticity

The property of solid to not regain its shape.



→ F_{ext}

~~$F_R = 0$~~



Stress

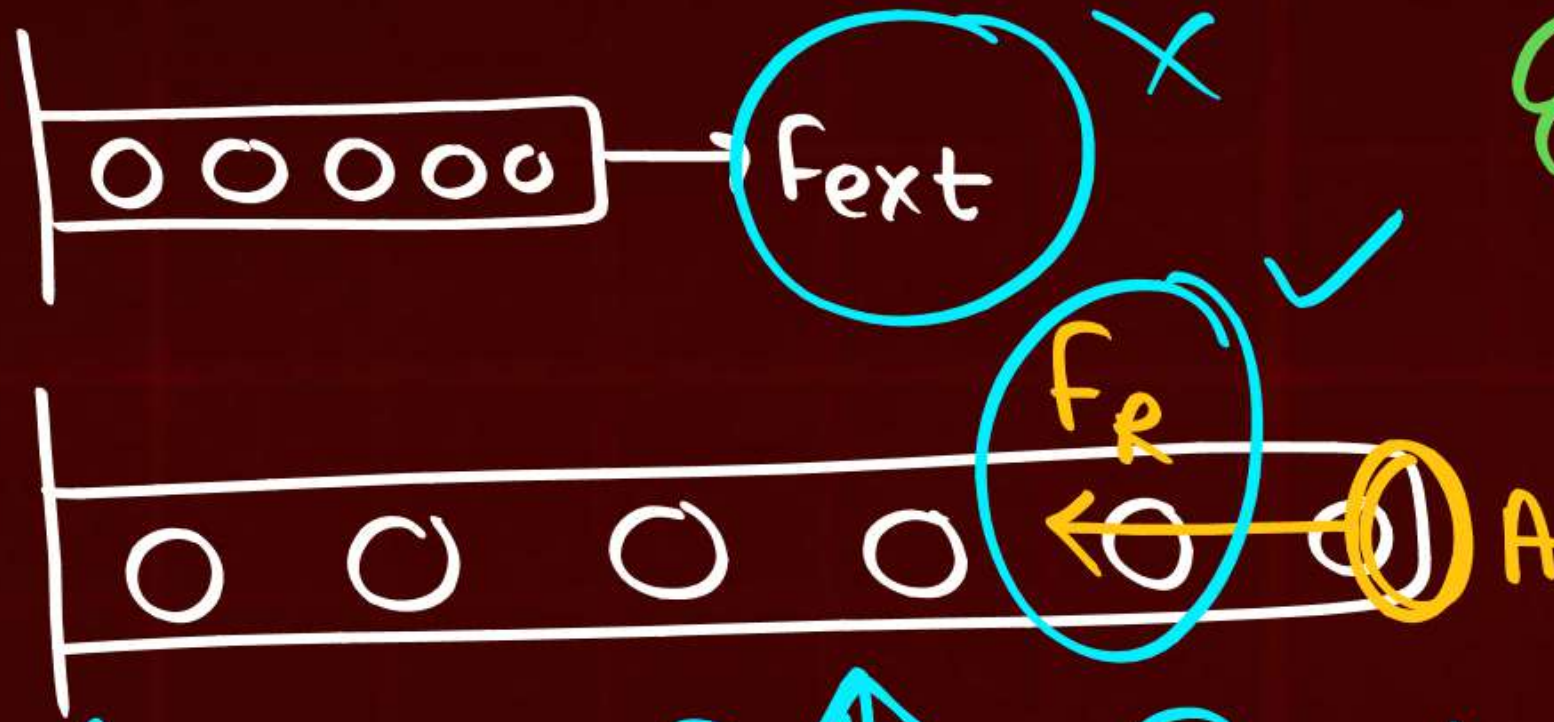
It is the restoring force acting per unit area.

$$\text{Stress} = \frac{F_R}{A} = \frac{N}{m^2} = \text{Pascal}$$

neet
2021

Dimensional

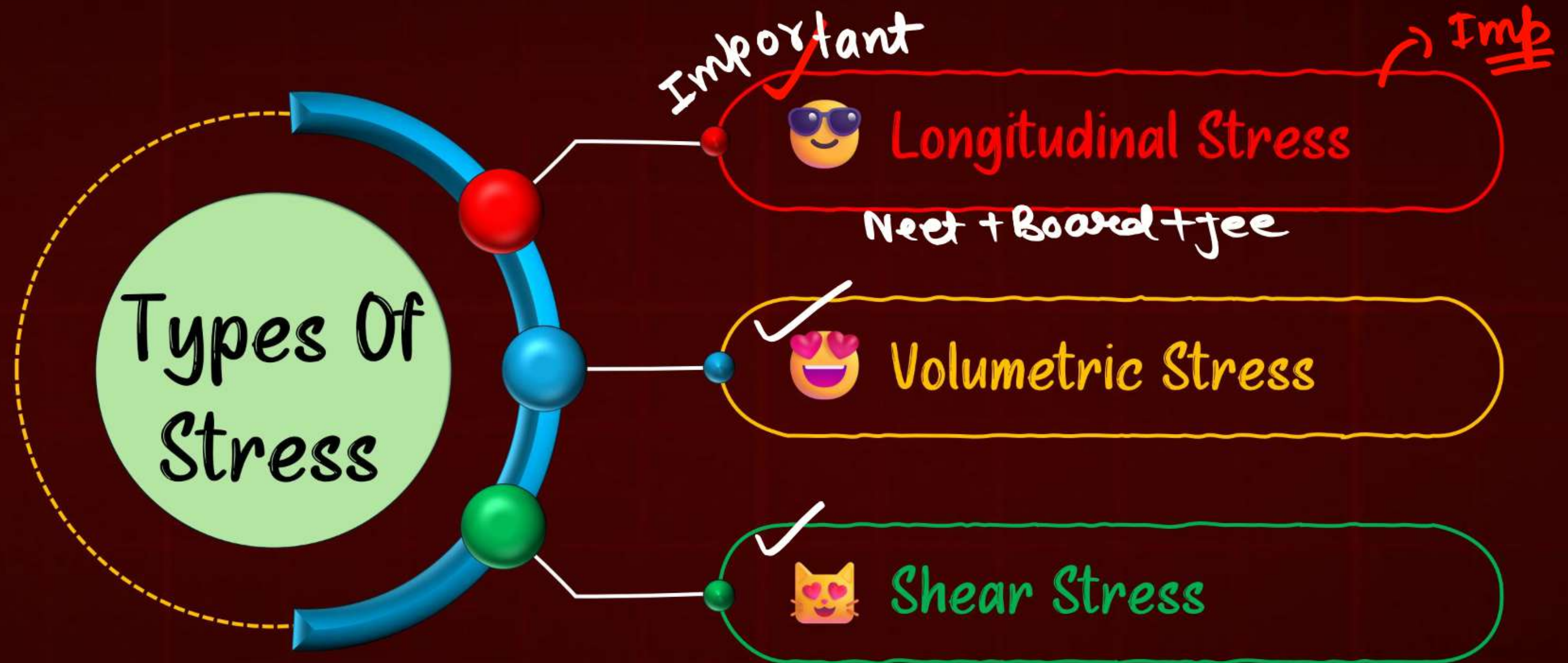
formula



$$\uparrow \text{Stress} = \frac{F_R}{A} \uparrow$$

$$\text{Elasticity} \uparrow \uparrow \uparrow = \frac{m \times T^{-2}}{L^2} = \underline{m L^{-1} T^{-2}}$$

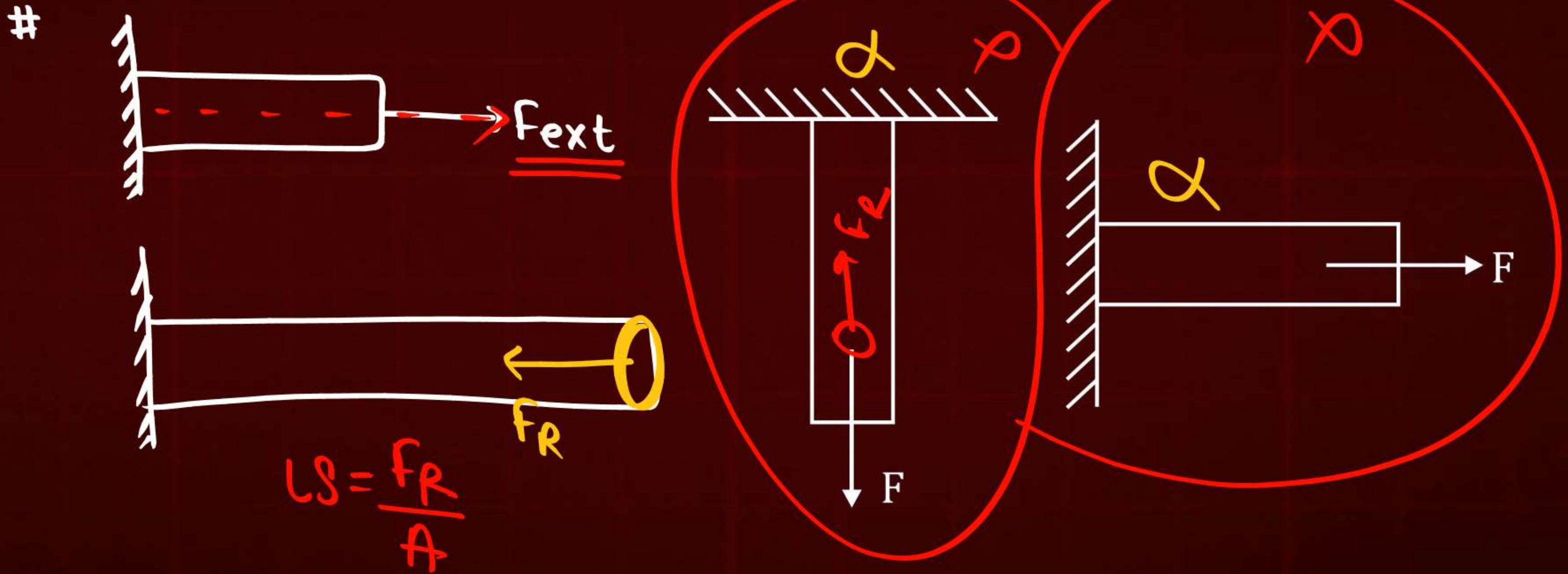
Types of Stress



Longitudinal Stress

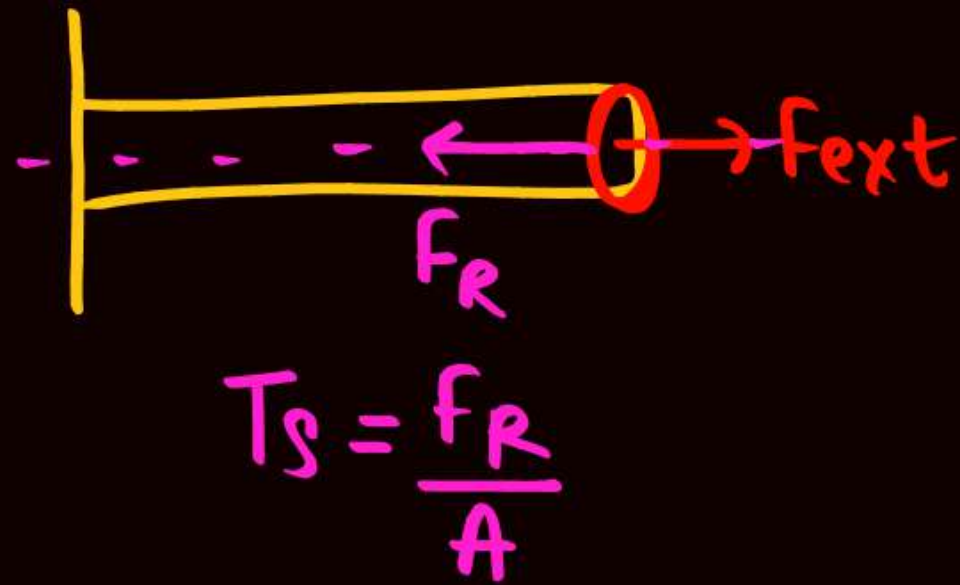
(Imp)

The stress developed when the force is applied along the Solid

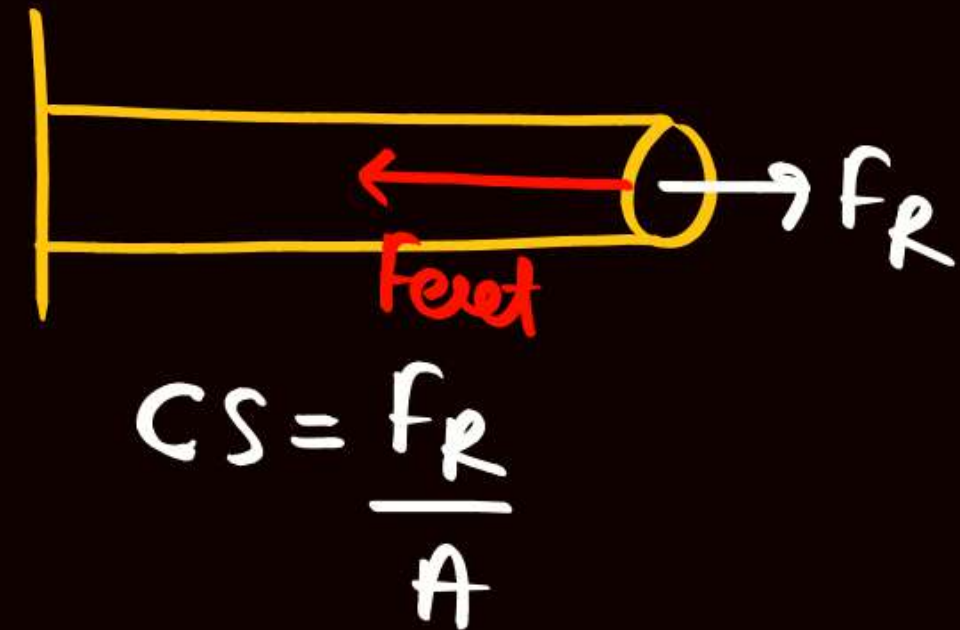


longitudinal stress

Tensile stress



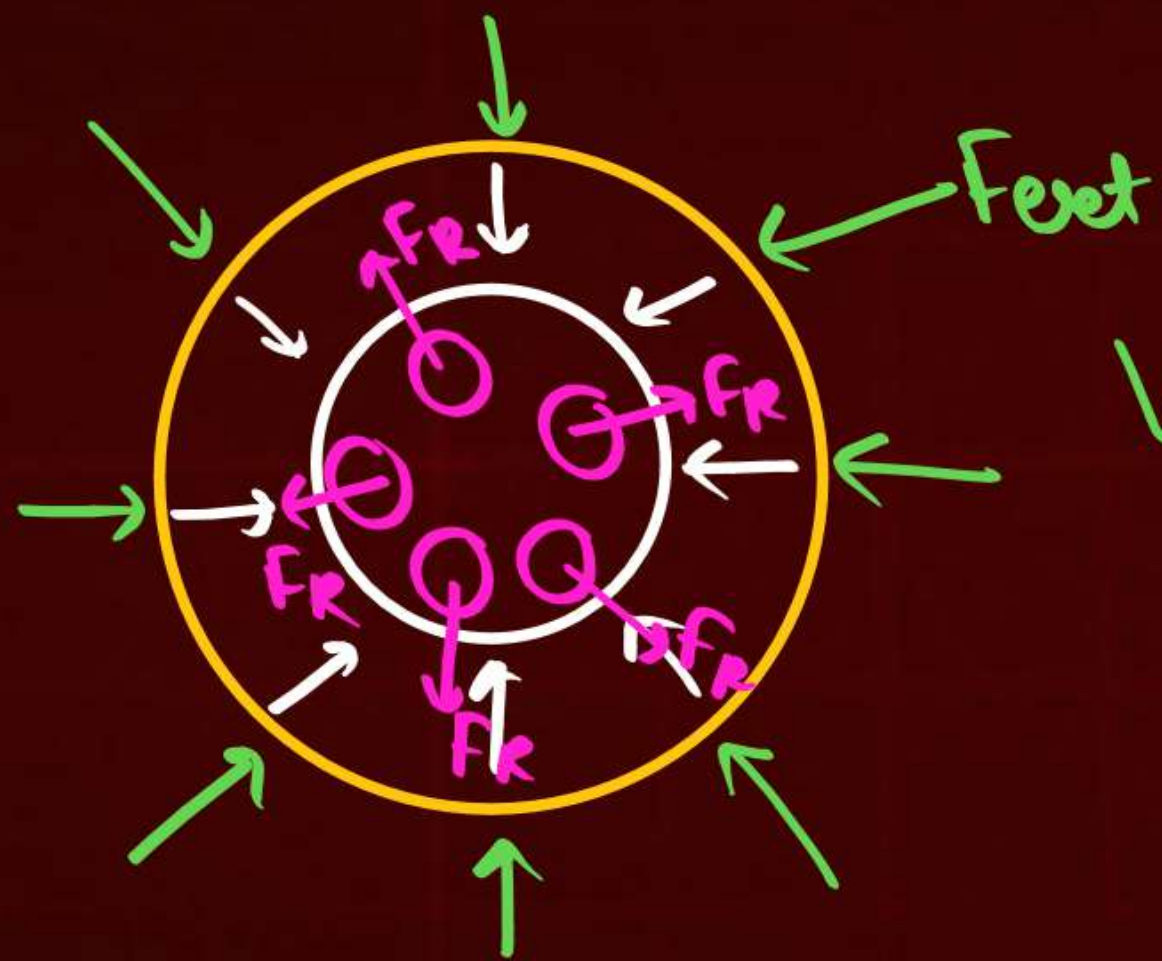
Compressive stress



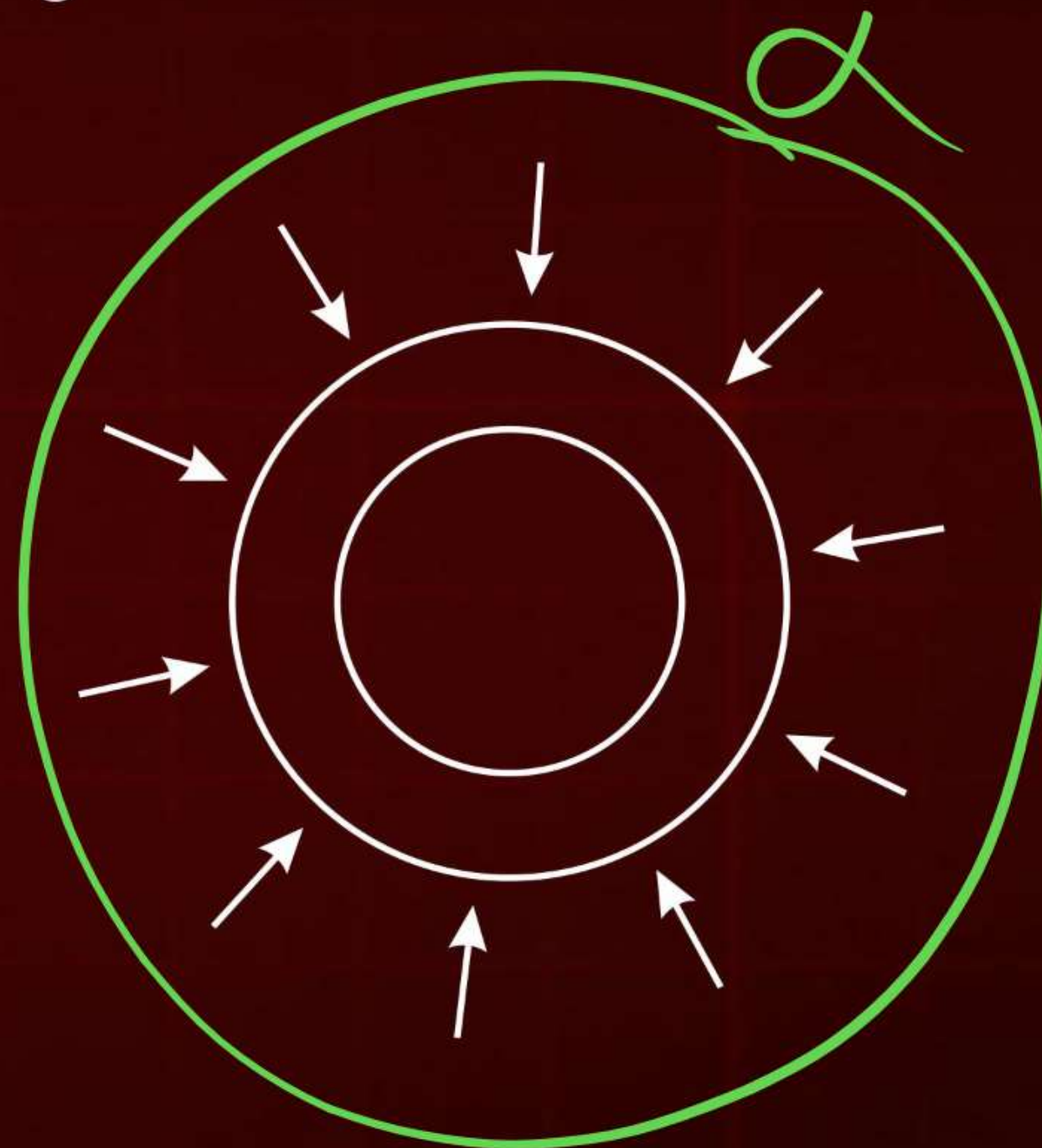


Volumetric Stress

The stress is developed when force is applied along the volume



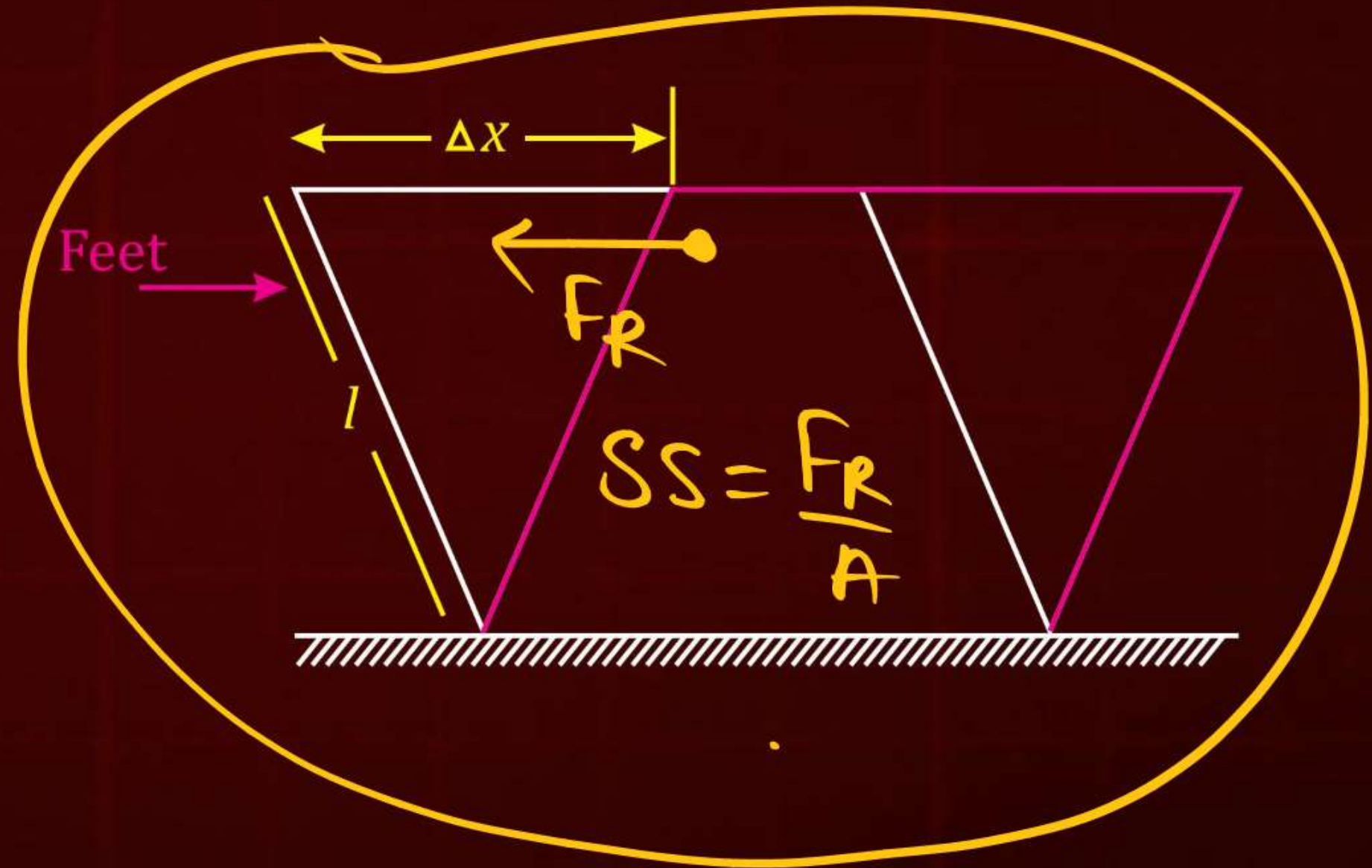
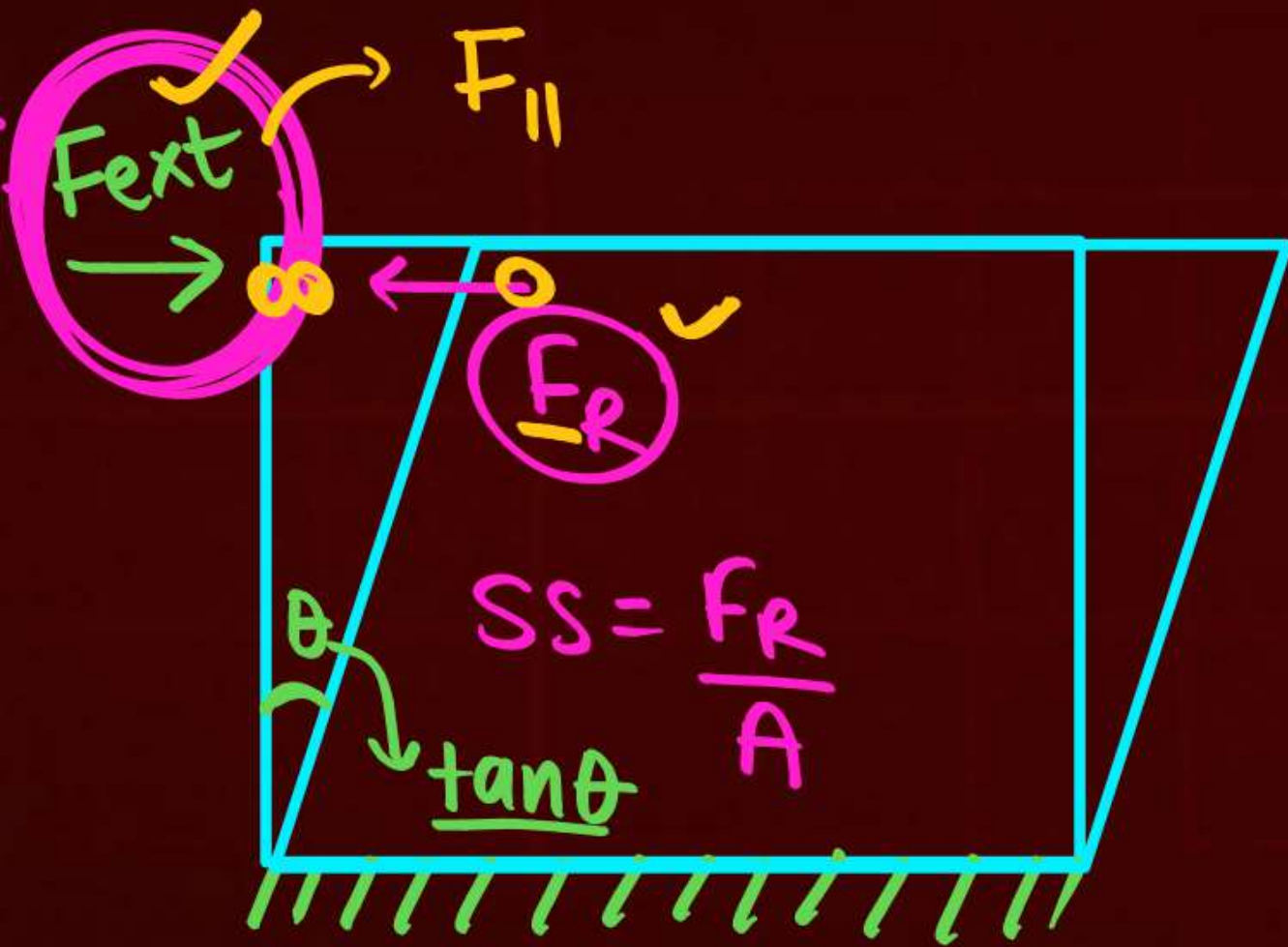
$$VS = \frac{fR}{A}$$



3

Shear Stress

The stress is developed when force is applied tangentially. tangentially ($F_{||}$)



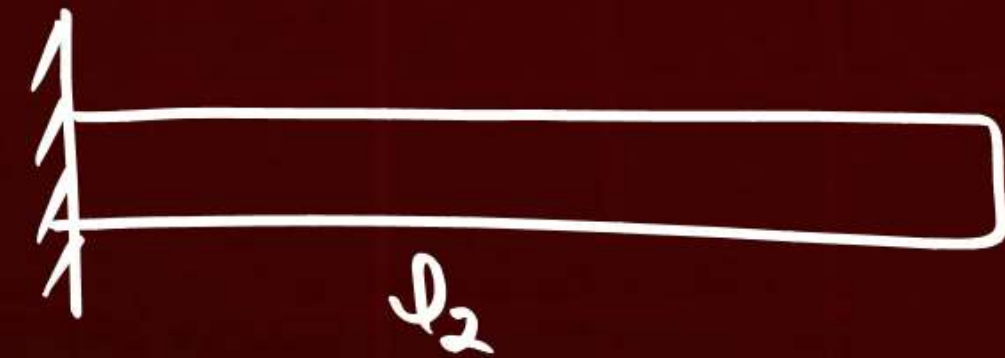
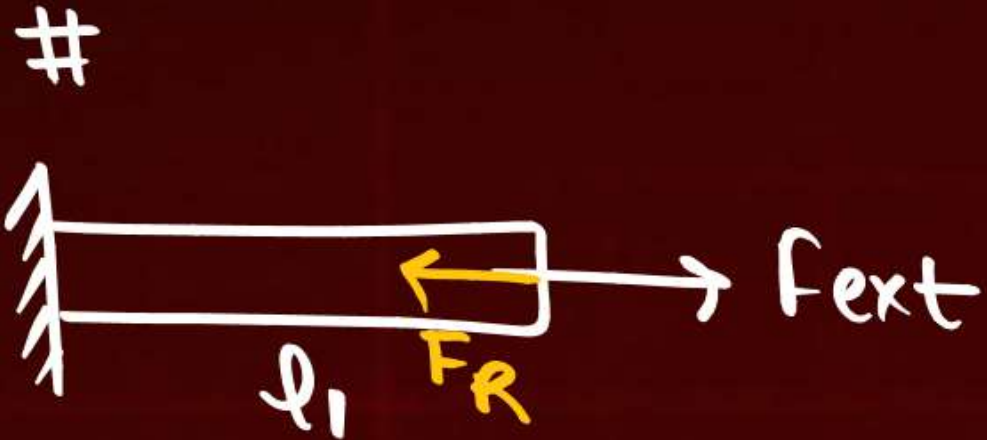
Q The Stress depend on

(a) Restoring Force

(b) F_{ext}

Strain

It is the ratio of change in ~~original~~ configurations to the initial original Configuration



$$\text{Strain} = \frac{\Delta \text{change in Configuration}}{\text{Original Configuration}}$$

$$\begin{aligned} \text{Strain} &= \frac{\Delta l}{l_1} \\ &= \frac{l_2 - l_1}{l_1} \end{aligned}$$

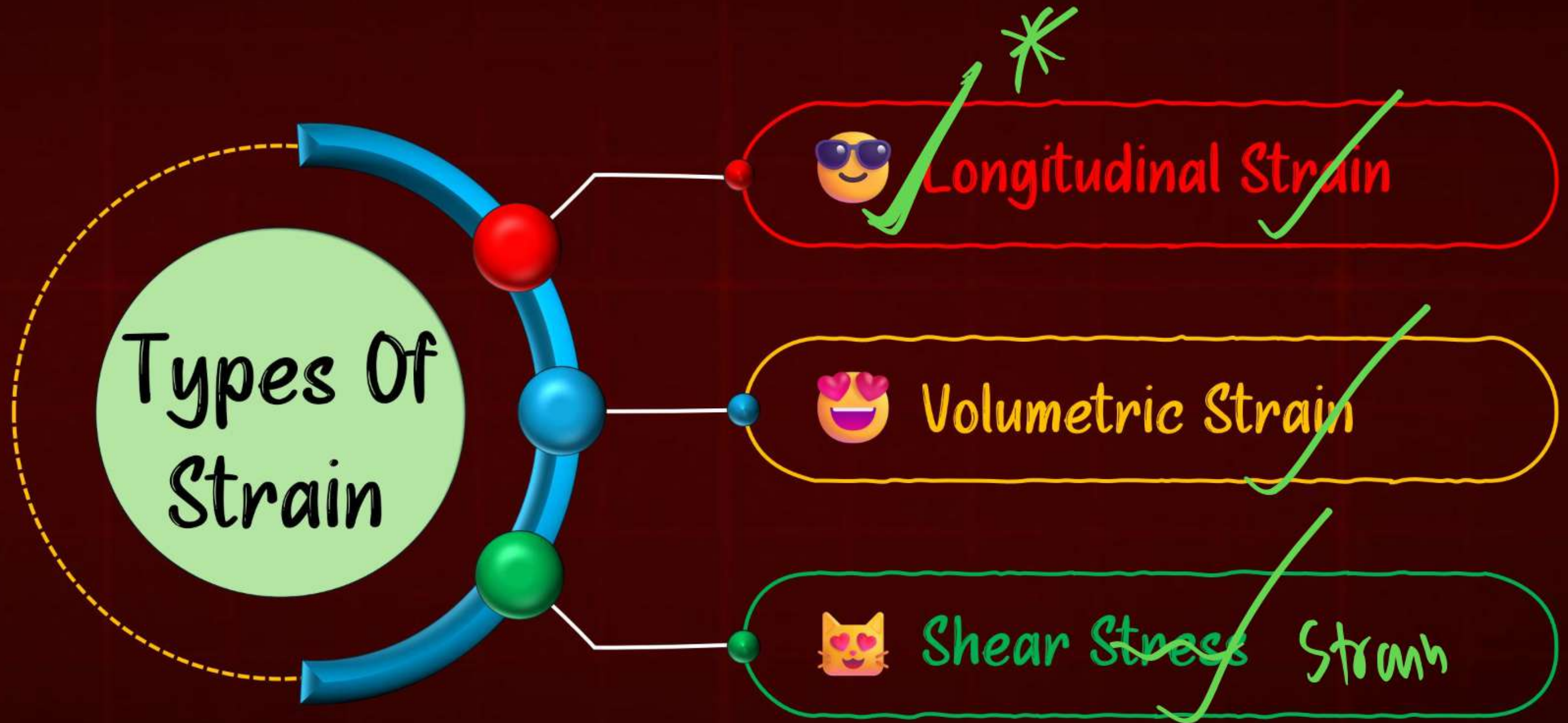
Strain only depend on F_ext

Unit of Strain

$$\text{Strain} = \frac{\Delta l}{l} = \frac{\cancel{\text{m}}}{\cancel{\text{m}}} = 1 \text{ Unitless}$$

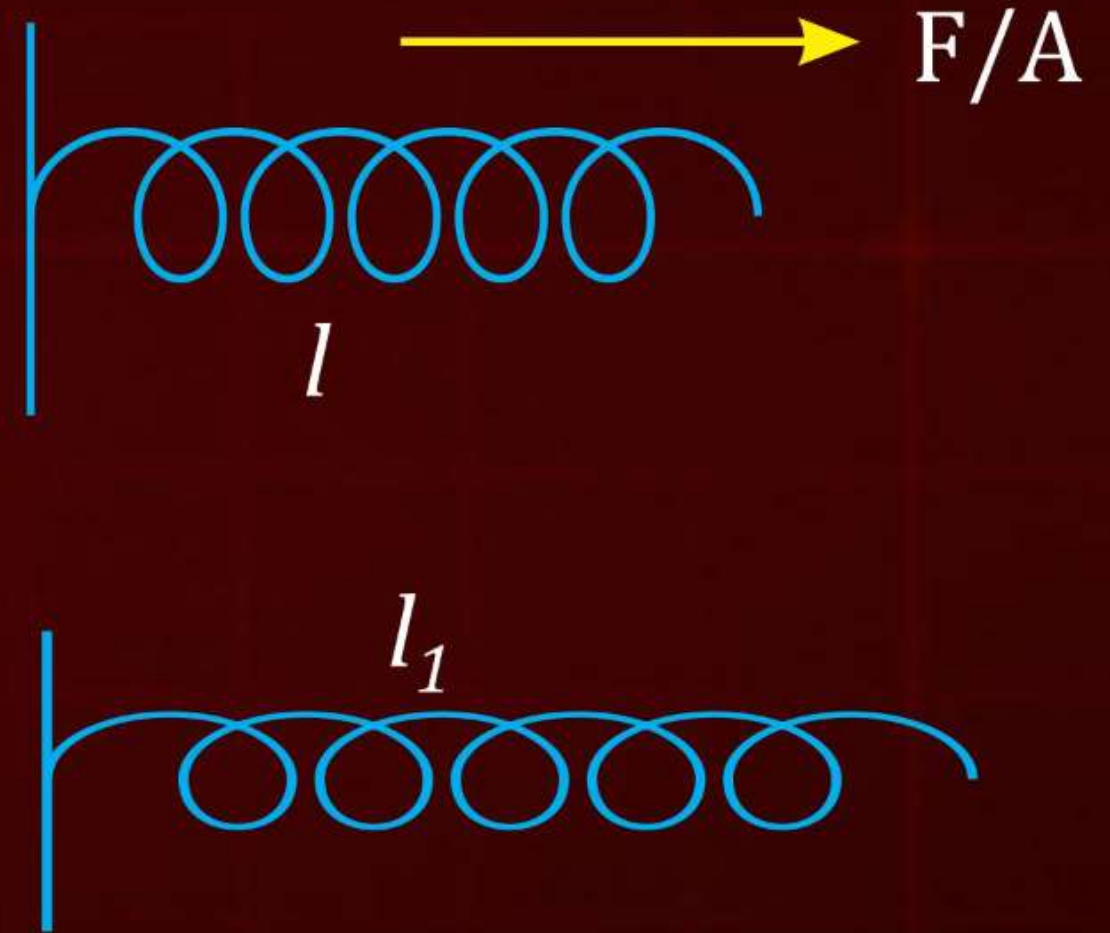
$$\text{Strain} = \frac{K}{x} = 1 \text{ (Dimensionless)}$$

Types Of Strain



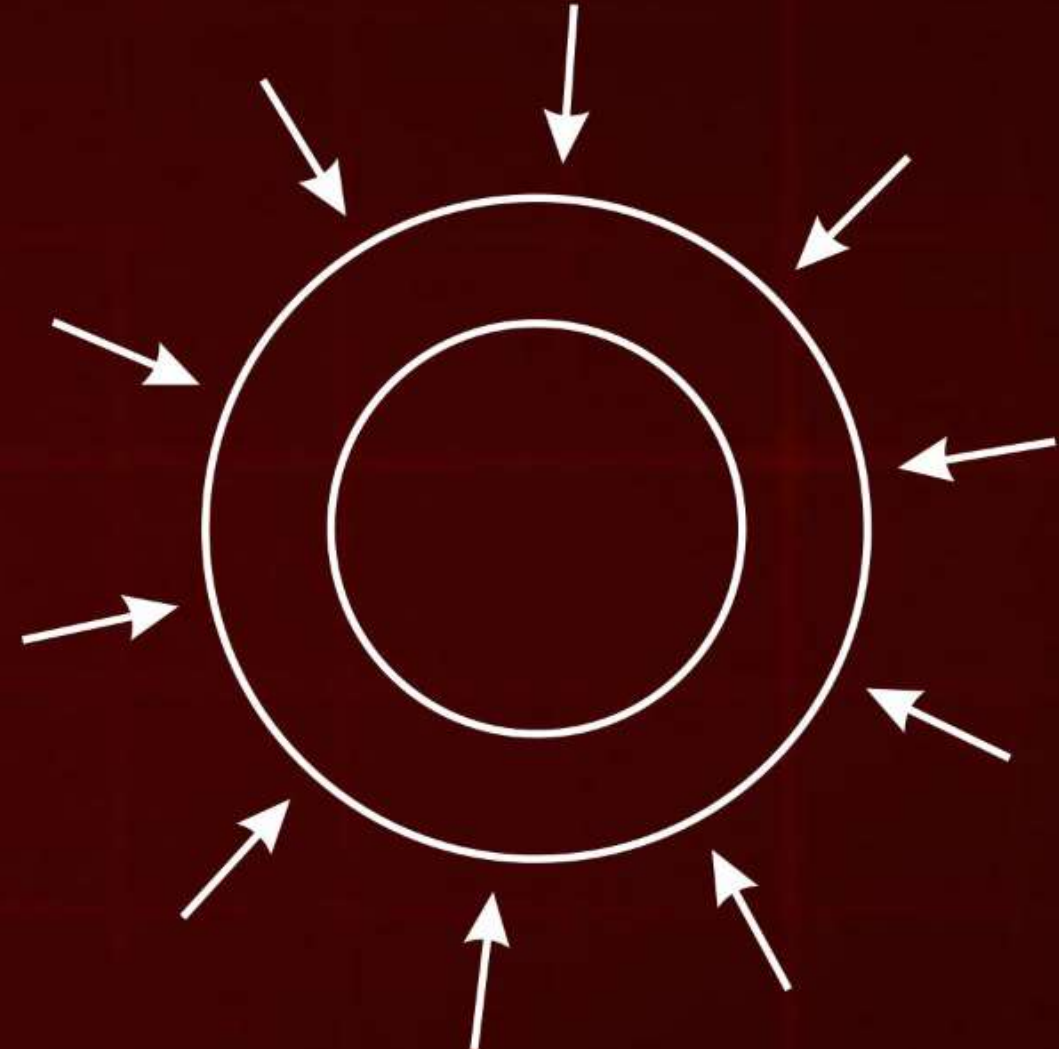
Longitudinal Strain

The strain caused by the longitudinal stress



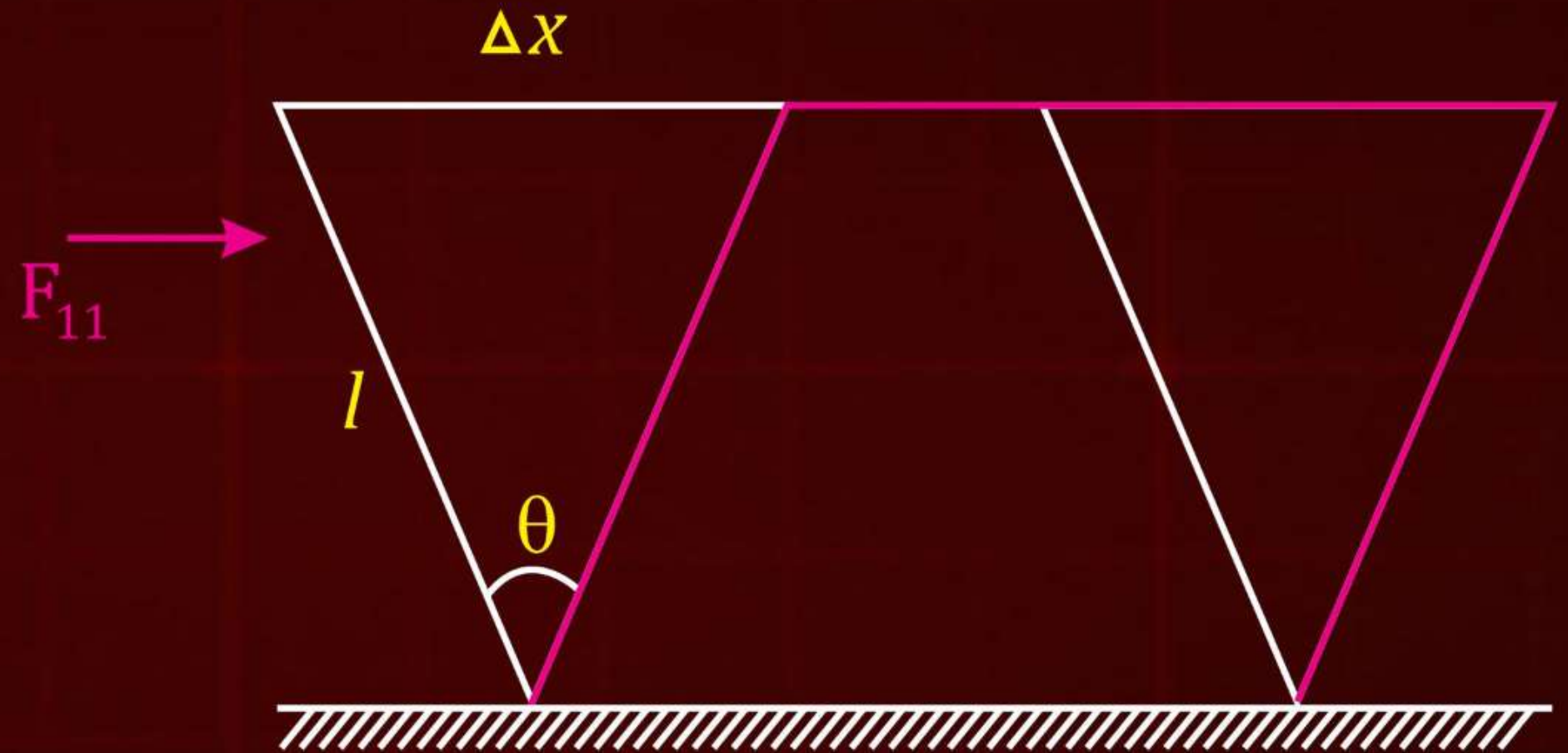
Volumetric Strain

The strain caused by volumetric stress



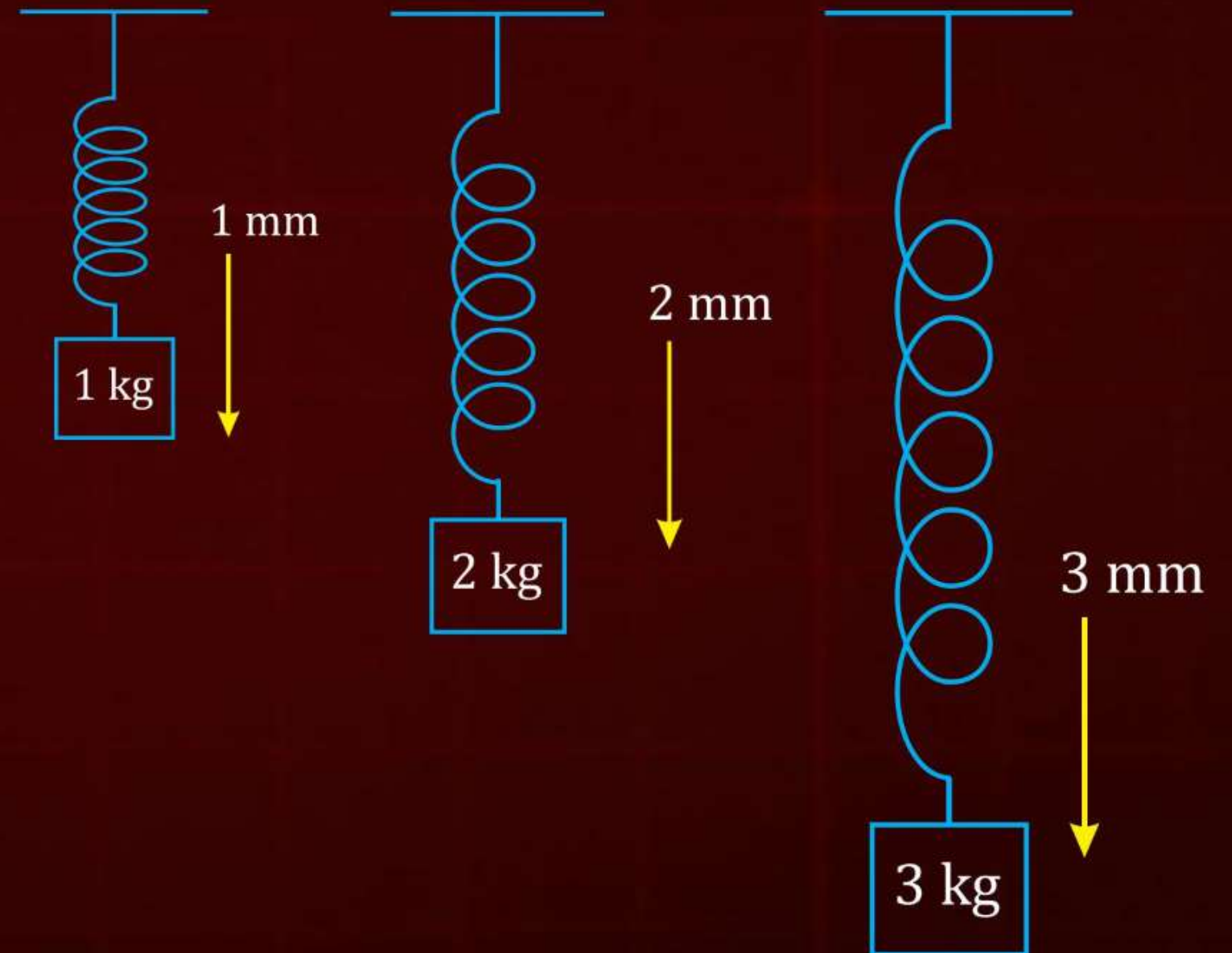
Shear Strain

The strain due to shear stress



Hookes Law

Hooker law states that the stress is directly proportional to strain



Modulus Of Elasticity (e)

1. Young modulus of e $= \frac{\text{Langitudinal Stress}}{\text{Longitudinal Strain}}$

2. Bulk modulus of e $= \frac{\text{Volumetric Stress}}{\text{V Strain}}$

3. Shear modulus of elasticity $= \frac{\text{Shear Stress}}{\text{Shear Strain}}$

Poisson Ratio

It is the negative of ratio of lateral strain to longitudinal stress

Elastic Potential Energy

The potential energy store when you stretch the rod.

Can you solve it?

✓ Stress is defined as:

$$\text{Stress} = \frac{F}{A}$$

$$S = \frac{F}{A}$$

$$= F \div A$$

- A** Force $\times$ Area
- B** Force $\div$ Area ✓
- C** Area $\div$ Force
- D** Mass $\div$ Volume

Answer

(B) Force $\div$ Area



Can you solve it?

The SI unit of stress is: (Neet-2022)

- A** Joule
- B** Newton
- C** Pascal ✓
- D** Watt

Answer

(C) Pascal



Can you solve it?

Strain is defined as the ratio of:

$$\text{Strain} = \frac{\Delta L}{L}$$

- A** Stress to force
- B** Change in length to original length
- C** Force to area
- D** Mass to volume

Answer

(B) Change in length to original length



Can you solve it?

The SI unit of strain is:

- A** Pascal
- B** Newton
- C** Meter
- D** ☒ No unit (dimensionless)

Answer

(D) No unit (dimensionless)



HOMework

• which one is more elastic

(a) Steel

(b) Iron

“Jab Ek Mard Padhta hai, Toh Sirf
ek Aadmi Padhta hai. Lekin Jab ek
Aurat Padh Jaati hai, Toh Poora
Khandaan Padh Jata hai.”



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